

Use Case: Library Book Issue & Return Service (REST-based)

Objective

Build a simple RESTful backend service to manage library books, members, and book issue/return operations using Java and Spring Boot.

Problem Statement

A college library wants to automate its book issuing process. The system should manage books, register members, issue books, and handle returns while enforcing basic business rules.

Actors

- Librarian
- Library Member

Core Data Objects

- Book – bookId, title, author, availability
- Member – memberId, name, email
- IssueRecord – issueId, issueDate, returnDate

Functional Requirements

Book Management

- Add a new book
- View all books
- View available books
- Search by title or author

Member Management

- Register a member
- View member details
- View books issued to a member

Issue Book

- Issue a book to a member
- Book must be available
- Member can issue a maximum of 3 books

Return Book

- Return an issued book
- Update return date
- Mark book as available

Basic Workflow

1. Add books
2. Register members
3. Issue books
4. Return books

Sample REST Endpoints

- POST /books
- GET /books
- GET /books/available
- POST /members
- POST /issues/issue
- PUT /issues/return/{issueId}

Business Rules

- One book can be issued to only one member at a time
- One member can have a maximum of 3 active book issues

Hackathon Expectations (2 Hours)

- Working REST APIs
- Validations and error handling
- Any database (H2/MySQL)

- Sample test data or Postman requests